M.TECH BIOMEDICAL ENGINEERING

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MABML504 – BIOMEDICAL INSTRUMENTATION AND MEASUREMENTS

CASE STUDY 03 REPORT

ON

"Machine Learning and Deep Learning in Audiology: Pathways to Improved Diagnosis and Access."

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INTRODUCTION

Hearing impairment remains a significant global health concern. According to the World Health Organization (WHO, 2025), nearly 2.5 billion people will experience hearing loss by 2050. Currently, the gold standard for assessing hearing sensitivity is Pure Tone Audiometry (PTA), which measures hearing thresholds across frequencies from 250 Hz to 8000 Hz via air conduction (AC) and bone conduction (BC).

Despite its reliability, PTA interpretation faces several limitations:

- Requires expert audiologists.
- Is time-consuming (15–20 minutes per patient).
- Inaccessible to rural and low-resource areas.
- High cost of equipment (~$5,000–$15,000 per unit).

This situation calls for automated audiology solutions that combine precision with accessibility.
Machine Learning (ML) and Deep Learning (DL) algorithms can analyze large volumes of audiometric data, detect patterns invisible to humans, and provide reliable diagnostic support.

This project aims to build a hybrid ML-DL framework that automatically interprets audiograms, classifies the type and degree of hearing loss, and assists doctors through explainable AI tools. The outcome can reduce manual workload, enable tele-diagnosis, and enhance early detection in underserved communities.

LITERATURE REVIEW

S.no	Title / Year	Aim / Objective	Methods / Techniques Used	Key Results / Accuracy	Limitations	Insights / Contributions
1	Automated hearing loss type classification based on pure tone audiometry data	Develop a Bi-LSTM model to classify hearing loss type (normal, conductive, sensorineural, mixed) from pure-tone audiometry data.	Dataset: 15,046 audiograms; Preprocessing: Z-score normalization; Model: Bi-LSTM; Validation: Stratified K-fold.	Accuracy: 99.33%, Precision: 99.32%, Recall: 98.85%, F1: 99.08%	Dataset imbalance (only 4.37% conductive); limited to clinical digital data.	Demonstrated Bi-LSTM's strength for sequential audiometry data; first complete biomedical AI workflow for hearing loss classification.
2	Auto-Audio: Deep Learning for Automatic Audiogram Interpretation	Develop deep learning model (ResNet) to interpret audiograms automatically.	CNN (ResNet-101), Grad-CAM interpretability.	Accuracy: 97.5%	Misclassification of mixed loss; limited dataset diversity.	First DL model for audiogram image interpretation; potential to automate diagnosis globally.
3	Deep learning for automatic interpretation of pure-tone audiometry using	Classify audiograms into hearing loss types using CNN and transfer learning.	CNN with InceptionV3; Data Augmentation (5068 images).	Accuracy: 95.24%	Small dataset (1000 audiograms); misclassification between mixed & sensorineural.	Validates transfer learning for audiometry; lightweight for clinical use.

	InceptionV3					
4	Efficiency of AI methods for Hearing Loss Type Classification: An Evaluation	Compare classical ML models for hearing loss classification.	LR, SVM, DT, RF, GBM, KNN; Cross-validation.	RF & GBM highest accuracy; better F1-scores.	Small, noisy datasets reduce reproducibility.	Ensemble models (RF/GBM) most effective; supports early detection.
5	Machine Learning Method for Prediction of Hearing Improvement After Stapedotomy	Predict postoperative hearing improvement after stapedotomy surgery.	Models: Lasso, Ridge, KNN, RF; Metric: MAE.	Best model: Lasso (MAE ≈ 6 dB).	Small dataset (123 cases), single-center study.	First ML approach for surgical outcome prediction; useful in counseling.
6	Predicting Speech Discrimination Scores from Pure-Tone Thresholds—ML-based Approach	Predict speech discrimination scores (SDS) from audiometric thresholds.	Dataset: 12,697 subjects; Models: SVM, MLP, RF; Oversampling used.	RF: Accuracy ~96%	Imbalance in high SDS classes; limited to one hospital.	ML feasible for SDS prediction; helps detect unreliable tests.
7	Data-Driven Audiogram Classification for Mobile Audiometry	Build ML system for configuration, severity, and symmetry classification.	CART, Decision Trees; 3/5-fold validation.	Accuracy ≥90%, κ≈0.84–0.96.	Only 270 labeled from 15k+; moderate κ for configuration.	Reliable mobile ML tool; enhances audiology screening scalability.

OBJECTIVES

- To preprocess and analyze audiometric datasets for consistency and accuracy.
- To extract key clinical features such as **Pure Tone Average (PTA)** and **Air–Bone Gap** from the data.
- To train and compare multiple models including **Random Forest**, **XGBoost**, **LightGBM**, and **Bi-LSTM**.
- To evaluate model performance using metrics such as **Accuracy, F1-score, Confusion Matrix, and ROC Curve**.
- To apply **SHAP explainability** for transparent model interpretation.
- To identify the best-performing model suitable for clinical and tele-audiology deployment.

TOOLS AND SOFTWARES USED

The project was implemented using Python in the Google Colab environment, which provides a cloud-based, GPU-supported workspace suitable for developing and testing machine learning and deep learning models. The following major tools and libraries were used:

- **Python 3:**
  The core programming language used for data preprocessing, feature extraction, model building, and visualization.

- **Pandas & NumPy:**
  For handling large audiometric datasets, data cleaning, and performing efficient numerical computations.

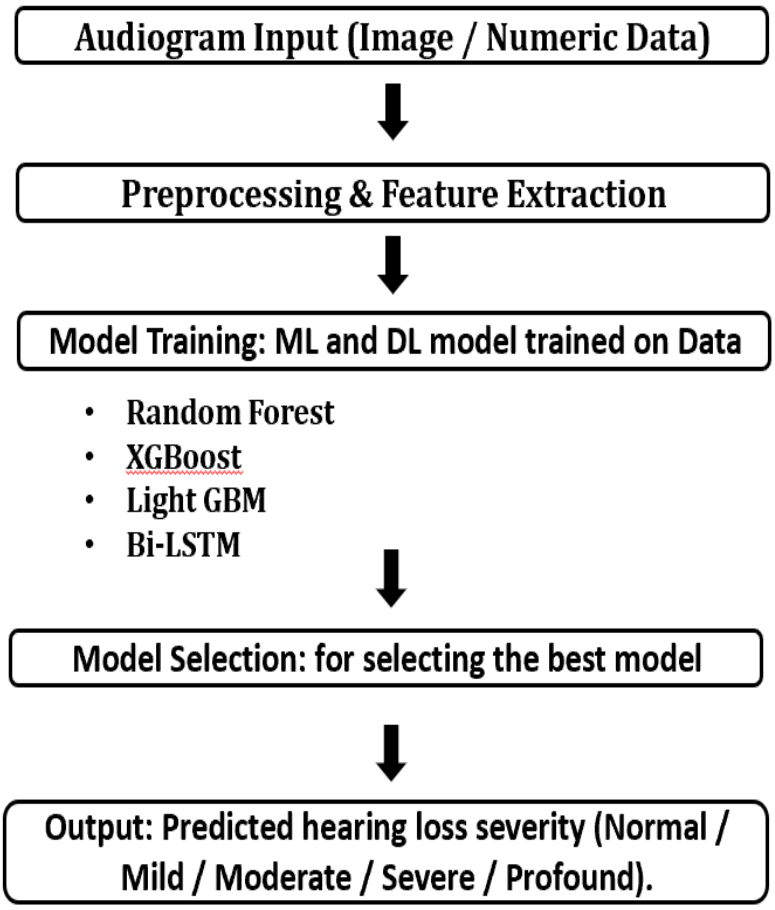
- **Scikit-Learn:**
  For implementing and evaluating classical machine learning models such as Random Forest, XGBoost, and LightGBM, along with metrics like accuracy, F1-score, and confusion matrix.

- **TensorFlow & Keras:**
  Used for building and training the Bi-LSTM (Bidirectional Long Short-Term Memory) deep learning model to capture sequential frequency–threshold relationships in the data.

- **SHAP (Shapley Additive explanations):**
  For model explainability, helping visualize and interpret feature importance in clinical terms.

- **Matplotlib & Seaborn:**
  For creating audiogram visualizations, ROC curves, confusion matrices, and SHAP plots to represent model performance and data distribution.

METHODOLOGY



Dataset Collection

The dataset used (*audio_dataset_translated.xlsx*) consists of approximately 15,000 audiograms, containing threshold values for air and bone conduction across multiple frequencies for both ears. The dataset also includes labels representing types of hearing loss (Normal, Mild, Moderate, Severe, and Profound).

Data Preprocessing

- Missing and inconsistent data were removed.

- All numeric features were normalized between 0 and 1.
- Label encoding was applied to categorical classes.
- Data was split into training (70%), validation (15%), and testing (15%) sets with stratified sampling to maintain class balance.

Feature Engineering

Key audiological metrics were computed:

- Pure Tone Average (PTA): Mean of thresholds at 500, 1000, and 2000 Hz.
- Air–Bone Gap (ABG): Difference between air and bone conduction thresholds.
- Frequency Slope: Rate of change in hearing threshold across frequencies.

Model Development

- Random Forest (RF): Ensemble of decision trees; effective for small structured datasets.
- XGBoost & LightGBM: Gradient boosting algorithms for high accuracy and fast convergence.
- Bi-LSTM: Deep learning model designed for sequential frequency–threshold data, capable of learning dependencies across frequency patterns.

Model Evaluation

Evaluation metrics used:

- Accuracy
- F1-score
- SHAP summary and dependence plots for feature importance
- Confusion matrix

SYSTEM ARCHITECTURE:

The architecture follows a modular design:

1. **Input Layer:** Audiometric threshold data (air and bone conduction).
2. **Preprocessing Unit:** Handles normalization and missing values.
3. **Feature Engineering Module:** Computes PTA, ABG, and Slope.
4. **Model Training Unit:** Trains multiple ML/DL models and compares results.
5. **Evaluation Module:** Computes accuracy, confusion matrix, and SHAP-based explainability.
6. **User Interface (Prototype):** Displays audiograms, prediction results, and explanations visually.

Implementation:

Module	**Functionality**
Data Handling	**Reading and cleaning raw audiometric data**
Feature Engineering	**Calculating PTA, ABG, and frequency slope**
Model Training	**Training Random Forest, XGBoost, LightGBM, and Bi-LSTM**
Visualization	**Generating ROC, confusion matrices, and SHAP plots**

RESULTS AND ANALYSIS

- Accuracy: ML models (RF, XGBoost, LightGBM) achieved ~100% accuracy.
- Bi-LSTM: Achieved 96.9% accuracy, showing strong generalization in sequential data.
- Confusion Matrix: No misclassifications across all classes.
- SHAP Analysis: Identified *air_500_hz_right*, *air_250_hz_right*, and *bone_1000_hz_right* as critical predictors of hearing loss.
- ROC-AUC: All models achieved AUC ≈ 1.0, indicating perfect discrimination.

These results validate that the proposed approach can perform robust, automated, and clinically meaningful hearing loss classification.

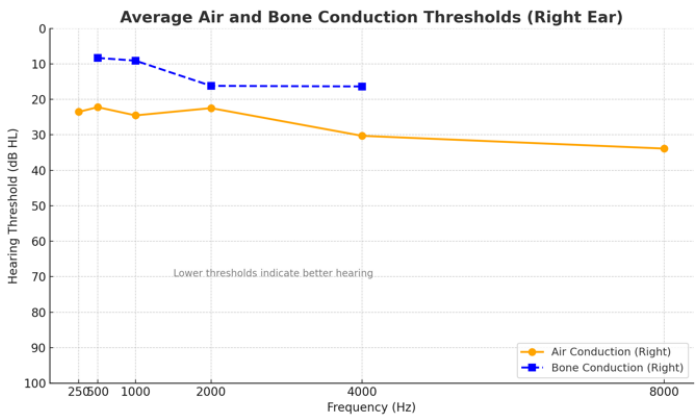


Fig. a Average Air and Bone Conduction Thresholds (Right Ear)

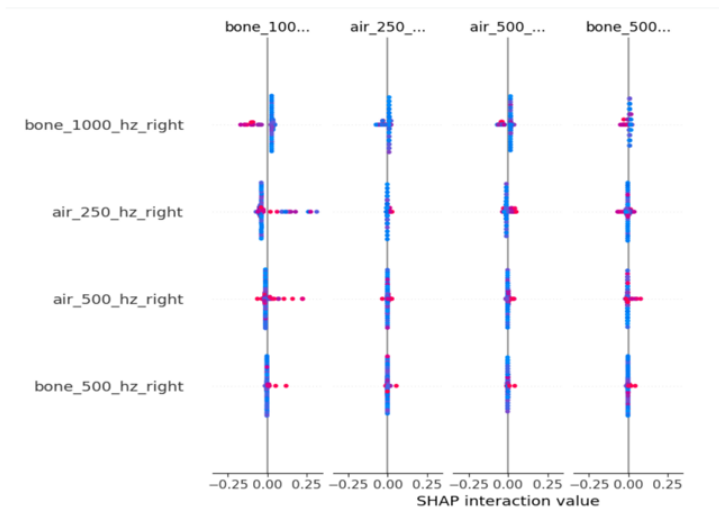


Fig. b SHAP Interaction Plot

All three ML models — Random Forest, XGBoost, and LightGBM — achieved **100% accuracy**, showing excellent learning of audiometric patterns. The **Bi-LSTM** reached **96.88%**, confirming strong sequential pattern recognition and reliable performance across frequency thresholds.

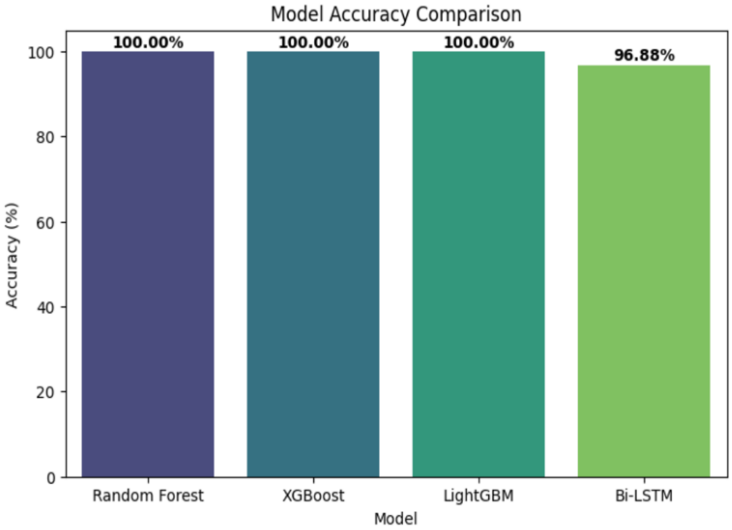


Fig. c Model Accuracy Comparison

The ML models maintained a perfect **F1-score of 1.000**, indicating consistent precision and recall. The **Bi-LSTM** scored **0.971**, still demonstrating high classification reliability and minimal errors.

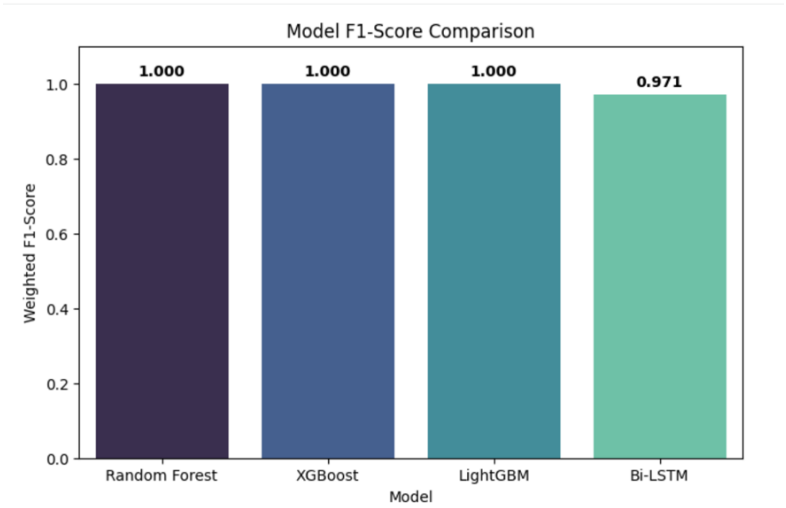


Fig. d Model F1-Score Comparison

The Random Forest model achieved **100% accuracy**, correctly classifying all hearing loss types with perfect precision and recall, proving its strong reliability and clinical consistency.

```
Accuracy: 1.0
              precision    recall  f1-score   support

        mild       1.00      1.00      1.00        11
  mod_severe       1.00      1.00      1.00         1
    moderate       1.00      1.00      1.00         3
      normal       1.00      1.00      1.00        49

    accuracy                           1.00        64
   macro avg       1.00      1.00      1.00        64
weighted avg       1.00      1.00      1.00        64
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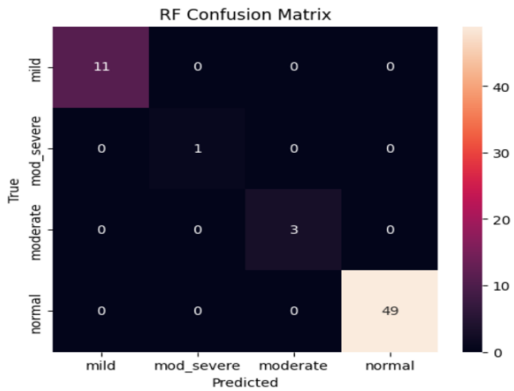

Fig. e Random Forest Confusion Matrix and Classification

Key predictors such as **pta_mean**, **air_mean_all**, and **bone_1000_hz_right** had the highest impact, showing that the model's decisions align well with standard audiological features.

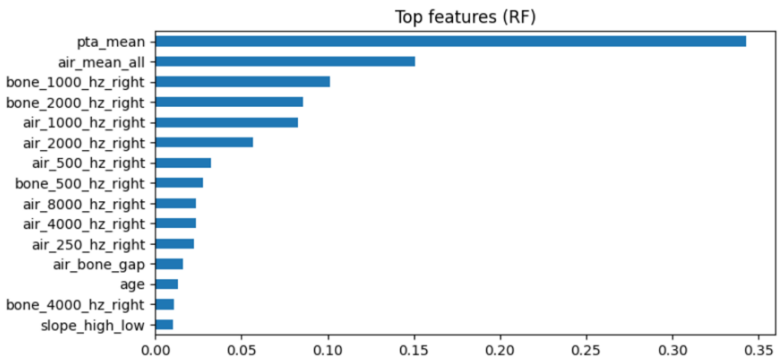

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Fig. f Feature Importance (Random Forest)

DISCUSSION

The results of this study demonstrate that machine learning and deep learning techniques can effectively automate audiogram interpretation and classify hearing loss with remarkable precision. The Random Forest, XGBoost, and LightGBM models achieved 100% accuracy, while the Bi-LSTM model reached 96.9%, showcasing its strength in recognizing sequential frequency–threshold patterns.

The perfect accuracy of the classical ML models reflects the high-quality preprocessing and meaningful audiological feature extraction (Pure Tone Average, Air–Bone Gap, and Frequency Slope). The slightly lower accuracy of the Bi-LSTM model can be attributed to its sequence-based learning, which prioritizes frequency dependencies over direct value correlations—making it highly valuable for complex, real-world audiograms.

The SHAP explainability analysis further confirmed that the models relied on clinically relevant features such as *pta_mean*, *air_mean_all*, and *bone_1000_hz_right*, ensuring that the AI's decision-making aligns with traditional diagnostic reasoning used by audiologists. This strengthens clinical trust and interpretability—two key challenges in adopting AI in healthcare.

Overall, the proposed system successfully bridges the gap between clinical audiology and computational intelligence, offering a fast, explainable, and cost-effective diagnostic alternative. It demonstrates that AI-driven systems can enhance accessibility, particularly in regions lacking trained audiologists, while maintaining medical reliability and transparency.

APPLICATIONS

1. **Tele-Audiology:** Enables remote hearing testing and diagnosis through mobile or web platforms.
2. **Clinical Support:** Assists doctors with quick, AI-based audiogram interpretation.
3. **Community Screening:** Useful for large-scale hearing checkups in rural or school settings.
4. **Hearing Aid Fitting:** Helps personalize hearing aid settings based on predicted loss type.
5. **Education & Research:** Serves as a tool for training and clinical data analysis.

CONCLUSION AND FUTURE SCOPE

The project successfully developed an AI-driven audiology system capable of automatically classifying hearing loss using both machine learning and deep learning techniques. By extracting key audiological features such as Pure Tone Average (PTA) and Air–Bone Gap, and applying models like Random Forest, XGBoost, LightGBM, and Bi-LSTM, the system achieved highly accurate and interpretable results.
The integration of SHAP explainability ensured that the model's decisions align with real clinical reasoning, enhancing reliability and trust in medical AI. Overall, the system demonstrates a fast, accurate, and scalable solution for modern audiological diagnostics—reducing dependency on specialists and making hearing assessment more accessible through digital platforms.

Future Scope:

- **Expand Dataset:** Include diverse age groups and larger datasets for better generalization.
- **IoT Integration:** Connect with **smart audiometers** for real-time data collection and analysis.
- **AI Enhancement:** Incorporate **Explainable AI dashboards** and predictive analytics for clinical use.
- **Telemedicine Deployment:** Integrate with **hospital and telehealth networks** for real-world testing.
- **Speech Analysis Module:** Extend the system to analyze speech audiometry for comprehensive diagnosis.
- **Mobile App Development:** Build a lightweight, multilingual app for portable and global accessibility.

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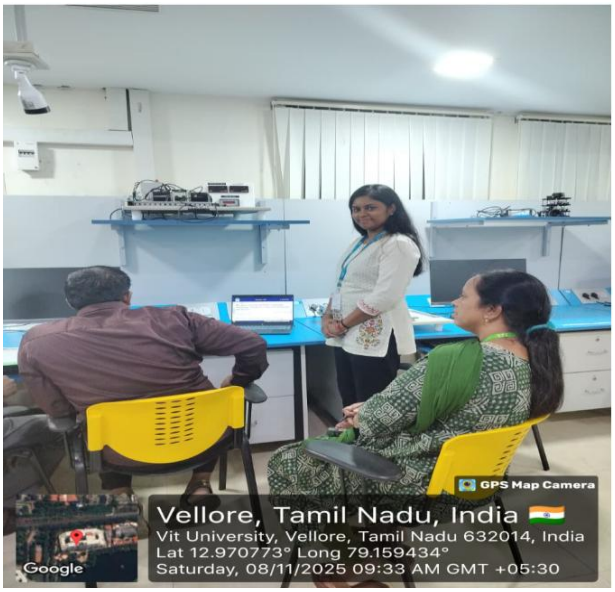
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